

## Intermediate Microeconomics — Week 3, Day 3

In-class Exercises: Repeated Games

work in pairs • solutions at the end

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Both exercises use the same stage game — a Prisoner's Dilemma between two firms choosing to **Cooperate** (keep price high) or **Defect** (cut price). Row = firm 1, payoffs in \$m per period:

|           | Cooperate | Defect |
|-----------|-----------|--------|
| Cooperate | (4, 4)    | (0, 6) |
| Defect    | (6, 0)    | (2, 2) |

### Exercise 1 — Finite repetition

[competency: backward induction and unraveling]

- (a) Find the Nash equilibrium of the one-shot stage game. Is Cooperate ever a best response?
- (b) The game is played exactly **three** times, and both firms know it. Using backward induction, determine what each firm does in period 3, then period 2, then period 1.
- (c) State the general lesson: for a stage game with a unique Nash equilibrium, what is the outcome of *any* finitely repeated version, and why does the *known endpoint* matter?

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### Exercise 2 — Infinite repetition and the critical discount factor

[competency: grim trigger, solve for  $\delta^*$ ]

Now the stage game is repeated **infinitely**, with common discount factor  $\delta \in (0, 1)$ . Each firm uses a **grim-trigger** strategy: cooperate until someone defects, then play Defect forever.

- (a) Write down the present value of (i) cooperating forever and (ii) defecting once and then being punished forever. Recall a constant stream of  $\pi$  per period is worth  $\pi/(1 - \delta)$ .
- (b) Derive the critical discount factor  $\delta^*$  above which cooperation is sustainable.
- (c) *Bigger temptation.* Suppose successful defection now pays 8 (the off-diagonal becomes (8, 0)/(0, 8)), punishment unchanged at 2. Recompute  $\delta^*$ . Does cooperation become easier or harder?
- (d) *Milder punishment.* Back to a defection payoff of 6, but now the punishment (both-Defect) payoff is 3 instead of 2. Recompute  $\delta^*$  and interpret.

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*Reconvene:* state your  $\delta^*$  for each case and, in one sentence, what makes tacit collusion *harder* to sustain.